Supplementary Appendix C - QUADAS-2 Risk of Bias Assessment

Abu-Alrub S, Strik M, Ramirez FD, Moussaoui N, Racine HP, Marchand H, Buliard S, Haïssaguerre M, Ploux S and Bordachar P (2022)

Domain 1 - Patient Selection

Describe methods of patient selection		Non-randomized: ● >18 years olf ● 100 patients in sinus rhythm who had undergone an AF ablation procedure in the previous 6 months ● 100 patients in persistent or permanent AF who were referred for catheter ablation
Was a consecutive or random sample of patients enrolled?	Consecutive	
Was a case-control design avoided?	Risk: High	
Did the study avoid inappropriate exclusions?	Risk: Low	Exclusion: ● Patients with atrial flutter, pacemakers or ICD
Did the selection of patients have introduced bias?	Risk: High	Case-control design was evident. Known SR patients were compared to patients with known AF. This does not represent the review population accurately as in the general population patients are mixed, symptoms vary and disease severity varies

Concerns regarding applicability		
Describe included patients		Location: Bordeaux. Their mean age was 62 ± 7 years and 56% were male.
Is there concern that the included patients do not match the review question?	Concern: Unclear	Patients post ablation may not represent the general population, however undergoing an ablation does not affect a device's ability to record ECG on the wrist.

Domain 2 - Index Test

Describe the index test and how it was conducted and interpreted	Low	After providing standardized instructions, 30 second recording of each ECG smartwatch, in random order Software version not noted
Were the index test results interpreted without knowledge of the results of the reference standard?	Risk: Low	
If a threshold was used, was it pre-specified?	Risk: Low	AF, SR, or unclassified (unable to differentiate between AF and SR).
Could the conduct or interpretation of the index test have introduced bias?	Risk: Low	

Concerns regarding applicability		
Is there concern that the index test, its conduct, or interpretation differ from the review question?	Concern: Unclear	The index test was performed in a controlled, clinical setting. This does not represent real-world use. This methodology is consistent throughout all studies, and is a general consideration for the outcome of this review.

Domain 3 - Reference Standard

Describe the reference standard and how it was conducted and interpreted		12-lead ECG collected immediately before the index test. Interpreted by 2 blinded electroophysiologists who independently interpreted each tracing and assigned one of three possible diagnoses: AF, SR, or unclassified. A third cardiac electrophysiology reviews the tracings in event of a disagreement.
Is the reference standard likely to correctly classify the target condition?	Risk: Low	
Were the reference standard results interpreted without knowledge of the results of	Risk: Low	

the index test?		
Could the reference standard, its conduct, or its interpretation have introduced bias?	Risk: Low	

Concerns regarding applicability		
Is there concern that the target condition as defined by the reference standard does not match the review question?	Concern: Low	The duration of the reference test was not stated.

Domain 4 - Flow and Timing

Describe any patients who did not receive the index test(s) and/or reference standard	Risk: Low	All patients received index test and reference standard
Describe the time interval and any interventions between index test(s) and reference standard	Risk: Low	Immediate
Was there an appropriate interval between index test(s) and reference standard?	Risk: Low	
Did all patients receive a reference standard?	Risk: Low	
Did patients receive the same reference standard?	Risk: Low	
Were all patients included in the analysis?	Risk: Low	Provided statistical analysis with all patients data (including “unclassified" index tests as false positives when the patient was in SR, and false negatives when the patient was in AF).
Could the patient flow have introduced bias?	Risk: Low	

Risk of Bias Summary			
Patient Selection	Index Test	Reference Standard	Flow and Timing

Applicability Summary		
Patient Selection	Index Test	Reference Standard

Mannhart D, Lischer M, Knecht S, du Fay de Lavallaz J, Strebel I, Serban T, Vögeli D, Schaer B, Osswald S, Mueller C, Kühne M, Sticherling C, Badertscher P (2023)

Domain 1 - Patient Selection

Describe methods of patient selection		April 2021 - May 2022 Location: Basel, Switzerland ● >18 years old ● Patients presenting to hospital for catheter ablation procedures, electrical cardioversions, pacemaker or ICD implantation.
Was a consecutive or random sample of patients enrolled?	Consecutive	
Was a case-control design avoided?	Risk: Low	There as no preestablished ‘healthy - SR” group vs “unhealthy - AF” group
Did the study avoid inappropriate exclusions?	Risk: Low	Exclusion ● Patients with 3 or less SL-ECG recordings due to logistical or technical issues or missing data
Did the selection of patients have introduced bias?	Risk: Low	

Concerns regarding applicability		
Describe included patients	Concern: Low	The median age was 67 (58-75) years, 113 (56%) subjects were 65

		years of age or older, and 31% were women. In 162 (81%) patients, there was a known history of AF. We identified 61 (30%) patients with AF (10 - [5%] patients with atrial flutter) at the time of the 12-lead and SL-ECG recording. A total of 133 (66%) patients were scheduled for pulmonary vein isolation, 26 (13%) for ablation of supraventricular tachycardia, 12 (6%) for ventricular tachycardia/premature ventricular complex ablation, 9 (4%) for cavotricuspid isthmus ablation, 5 (2%) for cardioversion, 8 (4%) for pacemaker implantation, and (4%) for implantable cardioverter-defibrillator implantation
Is there concern that the included patients do not match the review question?	Concern: Low	

<u>Domain 2 - Index Test</u>

Describe the index test and how it was conducted and interpreted	Risk: Low	4 different device brands were used. A 30 second recording from each of them was collected in a random order. All worn on the left wrist.
Were the index test results interpreted without knowledge of the results of the reference standard?	Risk: Low	
If a threshold was used, was it pre-specified?	Risk: Low	Sinus rhythm, AF, unclassified
Could the conduct or interpretation of the index test have introduced bias?	Risk: Low	

	Concerns regarding applicability	
	Concern: Unclear	The index test was performed in a controlled, clinical setting. This does not represent real-world use. This methodology is consistent throughout all studies, and is a general consideration for the outcome of this review.

Domain 3 - Reference Standard

Describe the reference standard and how it was conducted and interpreted	RIsk: Low	Schiller SDS – 200, Touch 4.4.3, Schiller 12-Lead ECG machine used to collect reference standards. Interpreted by 2 cardiologists. A third cardiologist reviewed any disagreements.
Is the reference standard likely to correctly classify the target condition?	Risk: Low	
Were the reference standard results interpreted without knowledge of the results of the index test?	Risk: Low	
Could the reference standard, its conduct, or its interpretation have introduced bias?	Risk: Low	

Concerns regarding applicability		
Is there concern that the target condition as defined by the reference standard does not match the review question?	Low	

Domain 4 - Flow and Timing

Describe any patients who did not receive the index	Risk: Unclear	10 patients were excluded from analysis due to incomplete index tests

test(s) and/or reference standard		
Describe the time interval and any interventions between index test(s) and reference standard	Risk: Unclear	“Index test followed reference standard”
Was there an appropriate interval between index test(s) and reference standard?	Risk: Unclear	Timeframe between index and reference was not specifically stated
Did all patients receive a reference standard?	Yes	
Did patients receive the same reference standard?	Yes	
Were all patients included in the analysis?	Yes	SL-ECGs labeled as inconclusive and the corresponding 12-lead cardiologist-interpreted ECGs labeled as AF were counted as false negatives; SL-ECGs labeled as inconclusive and the corresponding 12-lead cardiologist-interpreted ECGs labeled as sinus rhythm were counted as false positives.
Could the patient flow have introduced bias?	Risk: Unclear	The unspecified time between reference standard and index test leads to uncertainty. Rhythm changes can occur, with higher risk as the amount of time between the reference and index test increases. If a patient’s rhythm changed, their reference standard would no longer represent an accurate reference for the index test.

Risk of Bias Summary			
Patient Selection	Index Test	Reference Standard	Flow and Timing

Applicability Summary		
Patient Selection	Index Test	Reference Standard

Müller M, Hanssen TA, Johansen D, Jakobsen Ø, Pedersen JE, Aamot Aksetøy IL, Rasmussen TB, Hartvigsen G, Skogen V, Thrane G. (2024)

Domain 1 - Patient Selection

Describe methods of patient selection		Between Oct 2020-March 2022. ● Patients aged > 18 years ● Had undergone elective left-sided open-heart valve surgery.
Was a consecutive or random sample of patients enrolled?	Consecutive	
Was a case-control design avoided?	Risk: Low	There as no preestablished 'healthy - SR" group vs "unhealthy - AF" group
Did the study avoid inappropriate exclusions?	Risk: Low	Exclusion: ● Patients unable to participate due to serious postoperative complications
Did the selection of patients have introduced bias?	Risk: Low	

Concerns regarding applicability		
Describe included patients		Location: North Norway Mean age 68, 78.5% were male, 100% were caucasian. 18 patients had a history of AF.
Is there concern that the included patients do not match the review question?	Concern: Unclear	Patients post elective heart valve surgery may not represent the general population, however undergoing this procedure does not affect a device's ability to record ECG on the wrist.

Domain 2 - Index Test

Describe the index test and how it was conducted and		Software versions 6.0 to 8.4.1.

interpreted		A trained physiotherapist instructed all participants on how to perform ECGs by placing their index finger on the smartwatch's digital crown, keeping their arms rest on a table or legs, SL-ECG recordings from Apple Watch were recorded 3 times a day at specific times for 30 seconds. These were recorded on either the 2nd and 3rd or the 3rd and 4th post operative day.
Were the index test results interpreted without knowledge of the results of the reference standard?	Risk: Low	
If a threshold was used, was it pre-specified?	Risk: Unclear	No threshold stated
Could the conduct or interpretation of the index test have introduced bias?	Risk: High	Unclassified or uninterpretable SL-ECG were discarded. This will improve statistical accuracy results.

Concerns regarding applicability		
Is there concern that the index test, its conduct, or interpretation differ from the review question?	Concern: High	The index test was performed in a controlled, clinical setting. This does not represent real-world use. This methodology is consistent throughout all studies, and is a general consideration for the outcome of this review.

<u>Domain 3 - Reference Standard</u>

Describe the reference standard and how it was conducted and interpreted		Mortara XScribe 6, a standard 12-lead ECG machine took a 12-lead ECG on day 3 post operation
Is the reference standard likely to correctly classify the	Risk: Low	

Describe the reference standard and how it was conducted and interpreted		Mortara XScribe 6, a standard 12-lead ECG machine took a 12-lead ECG on day 3 post operation
target condition?		
Were the reference standard results interpreted without knowledge of the results of the index test?	Risk: Low	
Could the reference standard, its conduct, or its interpretation have introduced bias?	Risk: Low	

Concerns regarding applicability		
Is there concern that the target condition as defined by the reference standard does not match the review question?	Concern: Low	

<u>Domain 4 - Flow and Timing</u>

Describe any patients who did not receive the index test(s) and/or reference standard		105 enrolled ● 2 patients' surgeries were postponed - didnt revice index test or reference standard. ● 2 patients had serious postoperative complications - didint receive index test or reference standard ● 1 withdrew for personal reasons - didnt receive index test of reference standard ● 7 patients had missing data due to technical and administrative failure - received index test and reference standard but were not included in analysis Of all patients, 93 (100%) completed the first and second, and 92 (99%), 87 (93%),

		82 (88%), and 64 (69%) completed the third, fourth, fifth, and sixth Apple Watch ECG meas-urements, respectively
Describe the time interval and any interventions between index test(s) and reference standard	Risk: Low	Index test taken within 10 minutes of reference standard
Was there an appropriate interval between index test(s) and reference standard?	Risk: Low	
Did all patients receive a reference standard?	Risk: Low	
Did patients receive the same reference standard?	Risk: Low	
Were all patients included in the analysis?	No	• 7 patients had missing data due to technical and administrative failure • 1 patient due to only completing 2 successful index tests
Could the patient flow have introduced bias?	Unclear	There is lack of explanation as to why all patients did not complete all 6 SL-ECG recordings

Risk of Bias Summary			
Patient Selection	Index Test	Reference Standard	Flow and Timing

Applicability Summary		
Patient Selection	Index Test	Reference Standard

Briosa E Gala A, Sharp AJ, Schramm D, Pope MTB, Leo M, Varini R, Banerjee A, Win KZ, Kalla M, Paisey J, Curzen N, Betts TR (2025).

Domain 1 - Patient Selection

Describe methods of patient selection		Between Dec 2021 and Dec 2022. ● Documented history of cardiovascular disease in their electronic patient records, defined broadly as any clinically diagnosed disorder affecting the heart or blood vessels. ● Patients were eligible if they were undergoing a 12-lead ECG as part of routine clinical care
Was a consecutive or random sample of patients enrolled? Yes / No / Unclear	Consecutive	
Was a case-control design avoided?	Risk: Low	There as no preestablished 'healthy - SR" group vs "unhealthy - AF" group
Did the study avoid inappropriate exclusions?	No	Exclusion: ● Tattoos on the wrist
Did the selection of patients have introduced bias?	Risk: Low	

Concerns regarding applicability		
Describe included patients		Location: England The median age of participants was 66 years (interquartile range, 57– 75), and 29% were female. A history of AF was reported in 60.8% of patient, and 11.8% had a history of atrial flutter or AT. The presenting rhythm on the 12-lead ECG was AF in 194 (40%) of patients.
Is there concern that the included patients do not match the review question?	Concen: Low	This population represents the general population well, with a mixture of cardiovascular diseases, or differing severity.

<u>Domain 2 - Index Test</u>

Describe the index test and how it was conducted and interpreted		Algorithm version 1.0 SL-ECG from Apple Watch collected for 30

		seconds. If initial recording was deemed uninterpretable due to an artifact, a second recording was allowed. Two independent cardiologists adjudicated all SL-ECG recordings and 12-lead ECGs, with any disagreements resolved by a third cardiologist.
Were the index test results interpreted without knowledge of the results of the reference standard?	Risk: Low	
If a threshold was used, was it pre-specified?	Risk: Low	“AF,” “Not AF” (labeled as sinus rhythm by the Apple Watch), and “Unclassified.” For the Apple Watch, unclassified recordings were further categorized by the device as “Poor recording,” “Heart rate above 120 beats/min,” “Heart rate below 50 beats/min,” and “Inconclusive,” according to its internal criteria.
Could the conduct or interpretation of the index test have introduced bias?	Risk: Low	Unclassified SL-ECGs by the wearable devices were treated as false positives if the corresponding 12-lead ECG showed no AF, and as false negatives if the corresponding 12-lead ECG confirmed AF.

Concerns regarding applicability		
Is there concern that the index test, its conduct, or interpretation differ from the review question?	Concern: Unclear	The index test was performed in a controlled, clinical setting. This does not represent real-world use. This methodology is consistent throughout all studies, and is a general consideration for the outcome of this review. Unclassified SL-ECGs by the wearable devices were treated as false positives if the corresponding 12-lead ECG showed no AF, and as false negatives if the corresponding 12-lead ECG confirmed AF.

<u>Domain 3 - Reference Standard</u>

Describe the reference standard and how it was conducted and interpreted		Two independent cardiologists adjudicated all SL-ECG recordings and 12-lead ECGs, with any disagreements resolved by a third cardiologist.
Is the reference standard likely to correctly classify the target condition?	Risk: Low	
Were the reference standard results interpreted without knowledge of the results of the index test?	Risk: Low	
Could the reference standard, its conduct, or its interpretation have introduced bias?	Risk: Low	

Concerns regarding applicability		
Is there concern that the target condition as defined by the reference standard does not match the review question?	Concern: Low	

<u>Domain 4 - Flow and Timing</u>

Describe any patients who did not receive the index test(s) and/or reference standard		• 10 patients enrolled did not have optimal reference testing and were excluded from index test • 15 patients who were enrolled by ventricular pacing (exclusion criteria) did not receive the reference of the index test.
Describe the time interval and any interventions between index test(s) and reference standard	Risk: Low	Reference and index test done at the same time
Was there an appropriate interval between index test(s) and reference standard?	Risk: Low	

Did all patients receive a reference standard?	Risk: Low	
Did patients receive the same reference standard?	Risk: Low	
Were all patients included in the analysis?	No Risk: Low	Unclassified SL-ECGs by the wearable devices were treated as false positives if the corresponding 12-lead ECG showed no AF, and as false negatives if the corresponding 12-lead ECG confirmed AF.
Could the patient flow have introduced bias?	No	

Risk of Bias Summary			
Patient Selection	Index Test	Reference Standard	Flow and Timing

Applicability Summary		
Patient Selection	Index Test	Reference Standard

Osca Asensi J, Izquierdo de Francisco MT, Cano Pérez Ó, Sancho Tello de Carranza MJ, Alberola Rubio J, Planells Palop C, Lozano Vidal JV, Martínez Dolz L. (2021)

<u>Domain 1 - Patient Selection</u>

Describe methods of patient selection		Jan-June 2019 ● > 18 years old ● Referred to cardiology outpatients clinic for cardioversion of AF or for general consultation (patient with SR or AF).
Was a consecutive or random sample of patients enrolled?	Consecutive	

Was a case-control design avoided?	Yes Risk: Low	There as no preestablished ‘healthy - SR” group vs “unhealthy - AF” group
Did the study avoid inappropriate exclusions?	Risk: Low	Exclusion: ● Atrial flutter ● Pacemaker or ICD
Did the selection of patients have introduced bias?	Risk: Low	The addition of general outpatients made this study population a good representation of general population

Concerns regarding applicability		
Describe included patients		Location: Spain Valencia Mean age was 54, 63% male Hypertension was the most prevalent risk factor. 109 patients underwent cardioversion and 58 were seen for a general cardiology Consultation. The reference standard 12-lead ECG showed that 98 patients had AF and 69 had sinus rhythm
Is there concern that the included patients do not match the review question?	Concern: Low	

<u>Domain 2 - Index Test</u>

Describe the index test and how it was conducted and interpreted		The patient held their fingers to the two electrodes on the front of the watch for 3 minutes. Patients were instructed to sit while recording
Were the index test results interpreted without knowledge of the results of the reference standard?	Risk: Low	
If a threshold was used, was it pre-specified?	Risk: Unclear	No threshold stated

Could the conduct or interpretation of the index test have introduced bias?	Risk: High	167 total participants were involved, only 133 index tests were evaluated. No explanation was given as to why 34 patients were excluded. They do not state if unclassified or uninterpretable index test results were included or not.

Concerns regarding applicability		
Is there concern that the index test, its conduct, or interpretation differ from the review question?	Concern: Unclear	The index test was performed in a controlled, clinical setting. This does not represent real-world use. This methodology is consistent throughout all studies, and is a general consideration for the outcome of this review.

<u>Domain 3 - Reference Standard</u>

Describe the reference standard and how it was conducted and interpreted		Reference standard 12-lead ECG interpreted by 2 electrophysiologists was performed in all patients to determine whether they had sinus rhythm or AF.
Is the reference standard likely to correctly classify the target condition?	Risk: Low	
Were the reference standard results interpreted without knowledge of the results of the index test?	Risk: Low	
Could the reference standard, its conduct, or its interpretation have introduced bias?	Risk: Low	

Concerns regarding applicability

Is there concern that the target condition as defined by the reference standard does not match the review question?	No	

Domain 4 - Flow and Timing

	Yes / No / Unclear Risk: Low / High / Unclear	Details
Describe any patients who did not receive the index test(s) and/or reference standard	Risk: High	The study population was 167, but only 133 had index test data included in analysis. No justification as to why 34 were excluded.
Describe the time interval and any interventions between index test(s) and reference standard	Risk: Unclear	They state “after the 12-lead ECG, the [device] was placed on the patients wrists” No specific time frame stated
Was there an appropriate interval between index test(s) and reference standard?	Risk: Unclear	
Did all patients receive a reference standard?	Risk: Low	
Did patients receive the same reference standard?	Risk: Low	
Were all patients included in the analysis?	Risk: High	167 total participants were involved, only 133 index tests were evaluated. No explanation was given as to why 34 patients were excluded.
Could the patient flow have introduced bias?	Risk: High	

Risk of Bias Summary

Patient Selection	Index Test	Reference Standard	Flow and Timing

Applicability Summary		
Patient Selection	Index Test	Reference Standard

Wouters F, Gruwez H, Smeets C, Pijalovic A, Wilms W, Vranken J, Pieters Z, Van Herendael H, Nuyens D, Rivero-Ayerza M, Vandervoort P, Haemers P, Pison L. (2025)

Domain 1 - Patient Selection

Describe methods of patient selection		July-Nov 2023 ● Patients visiting the outpatient cardiology clinic were screened for eligibility ● All patients scheduled for cardioversion.
Was a consecutive or random sample of patients enrolled?	Consecutive	
Was a case-control design avoided?	Risk: Low	There as no preestablished ‘healthy - SR” group vs “unhealthy - AF” group
Did the study avoid inappropriate exclusions?	Risk: Low	Exclusions: ● Pacemaker ● Participation in another trial that would interfere ● Physical or cognitive restrictions
Did the selection of patients have introduced bias?	RIsk: Low	

Concerns regarding applicability		
Describe included patients	Concern: Low	The median age of the study population was 69 (IQR 61-77) years, with 44 of the 122 (36.1%) participants being women. All participants were White. Based on the 12-lead reference

		ECG, AF was present in 30 (24.6%) patients
Is there concern that the included patients do not match the review question?	Concern: Low	

Domain 2 - Index Test

Describe the index test and how it was conducted and interpreted		Doesn't state the series of Apple Watch used. To capture a lead-1 ECG using the Apple Watch, the participant wore the watch on the left wrist, ensuring skin contact with the electrode on the watch's underside. Measurement was initiated by the participant by touching a second electrode on the watch's crown with the right hand's index finger. Collected for 30 seconds
Were the index test results interpreted without knowledge of the results of the reference standard?	Risk: Low	
If a threshold was used, was it pre-specified?	Risk: Low	Sinus rhythm. High heart rate - no atrial fibrillation. Low or high heart rate. Atrial fibrillation. Atrial fibrillation - high heart rate, inconclusive, poor recording.
Could the conduct or interpretation of the index test have introduced bias?	Risk: High	Measurements of insufficient quality were repeated until the CWD algorithm provided a rhythm diagnosis. Good-quality measurements could not be obtained after three attempts in 4 patients. Measurements of insufficient quality were not included in analysis

Concerns regarding applicability

Is there concern that the index test, its conduct, or interpretation differ from the review question?	Concern: Unclear	The index test was performed in a controlled, clinical setting. This does not represent real-world use. This methodology is consistent throughout all studies, and is a general consideration for the outcome of this review.

Domain 3 - Reference Standard

Describe the reference standard and how it was conducted and interpreted		The reference diagnosis on the 12-lead ECG was provided by two cardiologists blinded to the device classifications and the participant's clinical information
Is the reference standard likely to correctly classify the target condition?	Risk: Low	
Were the reference standard results interpreted without knowledge of the results of the index test?	Risk: Low	
Could the reference standard, its conduct, or its interpretation have introduced bias?	Risk: Low	

Concerns regarding applicability		
Is there concern that the target condition as defined by the reference standard does not match the review question?	Concern: Low	

Domain 4 - Flow and Timing

Describe any patients who		200 patients enrolled. 17 removed

did not receive the index test(s) and/or reference standard		themselves, 60 not wanting to wait. 87 patients total did not receive the reference standard. 1 patient did not have an index test as their rhythm changed during the reference standard.
Describe the time interval and any interventions between index test(s) and reference standard	Risk: Low	12-lead and SL-ECG same time
Was there an appropriate interval between index test(s) and reference standard?	Risk: Low	
Did all patients receive a reference standard?	Risk: Low	All patients who were included in analysis underwent reference standard
Did patients receive the same reference standard?	Risk: Low	
Were all patients included in the analysis?	Risk: High	4 patients index test results were excluded due to insufficient quality
Could the patient flow have introduced bias?	Risk: High	“Insufficient” recordings were excluded, improving accuracy statistical analysis and reducing real-world application.

Risk of Bias Summary			
Patient Selection	Index Test	Reference Standard	Flow and Timing

Applicability Summary		
Patient Selection	Index Test	Reference Standard

Racine HP, Strik M, van der Zande J, Alrub SA, Caillol T, Haïssaguerre M, Ploux S, Bordachar P (2022)

Domain 1 - Patient Selection

	Yes/No/Unclear	Details
Describe methods of patient selection		● >18 years ● Admitted to cardiology department
Was a consecutive or random sample of patients enrolled?	Consecutive	
Was a case-control design avoided?	Risk: Low	There as no preestablished 'healthy - SR" group vs "unhealthy - AF" group
Did the study avoid inappropriate exclusions?	Risk: Unclear	No exclusions stated
Did the selection of patients have introduced bias?	Risk: Low	

Concerns regarding applicability		
Describe included patients		In total, 734 patients were enrolled with a mean age of 66 years, and 58% were male. Of these patients, 539 (73%) were in normal SR, 154 in AF (21%), 33 in atrial flutter (AFL) or atrial tachycardia (AT), 3 in ventricular tachycardia, and 5 in junctional tachycardia. Of note, 65 patients (8.9%) had sinus node dysfunction, 21 (2.9%) had second- or third-degree AV block, 39 (5.3%) had a ventricular paced rhythm, 54 (7.4%) had premature ventricular contractions (PVCs), and 132 (18%) had an intraventricular conduction delay (IVCD; right or left bundle branch block or nonspecific intraventricular conduction delay).
Is there concern that the included patients do not match the review question?	Concern: Low	

Domain 2 - Index Test

Describe the index test and how it was conducted and interpreted		Apple Watch Series 5 Patients received 5 standardized instructions to conduct index test Measurements taken for 30 seconds
Were the index test results interpreted without knowledge of the results of the reference standard?	Risk: Low	
If a threshold was used, was it pre-specified?	Risk: Low	The smartwatch's automated AF detection algorithm then concluded "no signs of AF," "AF," or "not checked for AF (unclassified)." Unclassified recordings were "low heart rate" (below 50 beats/min), "high heart rate" (above 150 beats/min), "poor recording," or "inconclusive recording."
Could the conduct or interpretation of the index test have introduced bias?	Risk: Low	unclassified ECGs were considered false positives (when the patient was in sinus rhythm [SR]) or false negatives (when the patient was in AF), yielding "worst-case scenario" estimates

Concerns regarding applicability		
Is there concern that the index test, its conduct, or interpretation differ from the review question?	Concern: Unclear	The index test was performed in a controlled, clinical setting. This does not represent real-world use. This methodology is consistent throughout all studies, and is a general consideration for the outcome of this review.

<u>Domain 3 - Reference Standard</u>

Describe the reference standard and how it was conducted and interpreted		12-Lead ECG interpreted by 2 cardiologists. A third cardiologist reviewed any disagreements.

Is the reference standard likely to correctly classify the target condition?	Risk: Low	
Were the reference standard results interpreted without knowledge of the results of the index test?	Risk: Low	
Could the reference standard, its conduct, or its interpretation have introduced bias?	Risk: Low	

Concerns regarding applicability		
Is there concern that the target condition as defined by the reference standard does not match the review question?	Concern: Low	

Domain 4 - Flow and Timing

Describe any patients who did not receive the index test(s) and/or reference standard	Risk: Unclear	No information
Describe the time interval and any interventions between index test(s) and reference standard	Risk: Low	SL-ECG done directly after 12-lead
Was there an appropriate interval between index test(s) and reference standard?	Risk: Low	
Did all patients receive a reference standard?	Risk: Low	
Did patients receive the same reference standard?	Risk: Low	
Were all patients included in	Risk: High	734 patients are stated to be

the analysis?		enrolled, however only 729 were mentioned in statistical analysis. Justification for exclusion was not stated.
Could the patient flow have introduced bias?	Risk: High	

Risk of Bias Summary			
Patient Selection	Index Test	Reference Standard	Flow and Timing

Applicability Summary		
Patient Selection	Index Test	Reference Standard

All Study Risk of Bias Summary				
	Patient Selection	Index Test	Reference Standard	Flow and Timing
Abu-Alrub S, 2022				
Mannhart D, 2023				
Müller M, 2024				
Briosa E Gala A, 2025				
Osca Asensi J, 2021				
Wouters F, 2025				
Racine HP, 2022				

All Study Applicability Summary			
	Patient Selection	Index Test	Reference Standard
Abu-Alrub S, 2022			
Mannhart D, 2023			
Müller M, 2024			
Briosa E Gala A, 2025			
Osca Asensi J, 2021			
Wouters F, 2025			
Racine HP, 2022			